

Kick-Off: Recent Developments in AI and Foundation Models

Statistical Programming
Prof. Dr. Martin Spindler

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Organization

Kick-Off Meeting



Seminar: Recent Developments in AI and Foundation Models

Winter Semester 2026/2027

Prof. Dr. Martin Spindler University of Hamburg

Welcome to the seminar kick-off!

Teaching Team

Prof. Dr. Martin Spindler

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- Office Hours: Wed 12:30 - 13:30

Supervisors

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- Valerian Fourel
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 - Moorweidenstrasse 18, Room 0025
- Julius Herzig
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Office Hours: on request via email

Seminar Timeline

1. **Review** the literature assigned to your topic
2. **Draft** an outline
3. **Discuss** your outline with your supervisor
4. **Write** the seminar paper and prepare the presentation
5. **Present** your work and actively participate in discussions



Warning

- Your seminar paper outline should be completed by **15th September** and agreed upon with your supervisor.
- Submission Deadline for the seminar paper is the **27th November, 23:59 CET**.

Important Dates

Statistical Programming

Date	From	To	Room	Lecturer
Thu, Jul 9, 2026	17:00	18:00	Digital	Prof. Dr. Martin Spindler
Fri, Dec 11, 2026	16:00	20:00	Mo 18, 0005.	Prof. Dr. Martin Spindler
Sat, Dec 12, 2026	10:00	17:00	Mo 18, 0005.	Prof. Dr. Martin Spindler
Sun, Dec 13, 2026	10:00	17:00	Mo 18, 0005.	Prof. Dr. Martin Spindler

All dates will be announced in time via STiNE and email.

Formal Requirements - Seminar Paper

- **Submission:** One printed copy (ring, thermal, or plastic binding) and one electronic copy (PDF), including code and simulations
- **Length:** 12-15 pages of main text
- **Typesetting:** Justified text
- **Printing:** Single-sided or with odd page numbers on the right side
- **No** further requirements regarding margins, font size, line spacing, or typeface

Recommendations - Seminar Paper

- Use **LaTeX** or **Quarto Markdown**
- Proportional font, e.g., Times New Roman
- Font size 11-12pt
- Line spacing 1.0 to 1.5

Note

Additional rules (including citation guidelines), tips, and templates: <https://www.bwl.uni-hamburg.de/statistik/studium/abschlussarbeiten.html>

Recommendations - Seminar Presentation

- Use **Quarto** or **PowerPoint**
- At least **2-3 minutes per slide**
- Typeface: sans-serif (e.g., Arial, Verdana)
- Minimum font sizes:
 - **18pt** for headings
 - **14pt** for subtitles, text, and numbers
 - **10pt** for indices and exponents
- Maximum **13 lines per slide**
- Present simulation results as **figures**

Topic Proposals

Suggested Topics



The following topic proposals are available for the seminar. Topics will be assigned via STiNE after the introductory meeting.

Topics 1 and 2

1. Medical Evidence Agent

Supervisor: Julius Herzig

In medicine, there are very structured ways in which results are presented. The agent should search for the evidence base and present it in a structured format accessible to clinicians.

Reference: Medical reporting standards - <https://www.equator-network.org/reporting-guidelines/consort/>

2. Gender Bias for Medical Topics

Supervisor: Julius Herzig

There is research that LLMs are biased in their answers for women compared to men. Investigate, for example, how pain is interpreted by LLMs depending on gender.

Reference: <https://www.jmir.org/2024/1/e54242/>

Topics 3 and 4

3. Agentic AI for Replicating Empirical Research

Supervisor: Martin Spindler

Reproducing published research is slow manual work that agentic AI is now capable of automating. Build and evaluate an agent, or a proof of concept, that runs replication packages from a small sample of papers.

Reference: Xu, Y., & Yang, L. Y. (2026). *Scaling reproducibility: An AI-assisted workflow for large-scale reanalysis*.

4. Jailbreaking Attacks on Large Language Models: Methods and Defense Strategies

Supervisor: Martin Spindler

LLMs face a growing risk of attacks that can make them produce harmful content or take unsafe actions. Review current attack methods and defense strategies.

Reference: Cunningham, H. et al. (2026). *Constitutional Classifiers++: Efficient Production-Grade Defenses against Universal Jailbreaks*.

Topics 5 and 6

5. SLM-Based Agent Conversion

Supervisor: Jan Rabenseifner

Replace parts of LLM-based agents with specialized SLMs to reduce cost and latency. Fine-tune a small model for a tool-calling task and compare it to a larger model. The goal is to critically assess the trade-offs between efficiency and capability, exploring when and how SLMs can replace LLMs in production-grade agentic systems without sacrificing functional adequacy.

References: - <https://arxiv.org/abs/2506.02153> - <https://github.com/ai-in-pm/Small-Language-Model-SLM-Guide>

6. LLMs as Simulated Economic Agents

Supervisor: Jan Rabenseifner

Use LLMs as agent-based simulations of human economic decision-making. The goal is to critically assess the validity, limitations, and scalability of this approach, and its potential as a complementary tool for economic research. **References:** -

<https://arxiv.org/abs/2301.07543> - Replicate a classic experiment with the

Topics 7 and 8

7. From Paper to Dataset: Agentic AI for Dataset Assembly

Supervisor: Erik Wendt

Build and evaluate LLM-based agents that read an empirical finance paper and autonomously assemble the corresponding dataset from WRDS.

Reference: <https://wrds-www.wharton.upenn.edu/pages/grid-items/using-python-wrds-platform/>

8. Benchmarks for Financial LLM Agents

Supervisor: Erik Wendt

Benchmarks for LLM agents in finance have grown rapidly but remain fragmented and hard to compare. This review organizes them along key dimensions and outlines directions for future evaluation.

Reference: <https://arxiv.org/abs/2402.12659>

Topics 9 and 10

9. Adversarial Attacks and Robustness

Supervisor: Jan Teichert-Kluge

Adversarial attacks are deliberate manipulation techniques designed to deceive machine learning models and artificial intelligence systems into making incorrect predictions, classifications, or decisions.

10. Uncertainty in Foundation Models

Supervisor: Jan Teichert-Kluge

Entropy of output utterances is high when there is syntactic diversity, whereas the real focus is semantic diversity.

References: - <https://www.nature.com/articles/s41586-024-07421-0> - <https://uncertainty-foundation-models.github.io/> - <https://www.nature.com/articles/s41467-025-64249-6>

Topic 11 and 12

11. Do Thumbnails Match Descriptions? Cross-Modal Embedding Alignment on YouTube

Supervisor: Valerian Fourel

Thumbnail and description of the same video should convey the same content. Embed both with vision-language models such as CLIP, learn a mapping between the two embedding distributions, and quantify how well they fit together, for example to flag clickbait or mismatched videos or to relate alignment to engagement.

References: - <https://arxiv.org/abs/2103.00020> - <https://arxiv.org/abs/2405.07987>

12. How Much of a Large Model Can You Distill into a Small One?

Supervisor: Valerian Fourel

Knowledge distillation transfers capabilities from a large teacher into a much smaller student. Review the distillation literature, including logit-, sequence-, and on-policy KD, and quantify how much capability survives at different teacher-student size ratios using only free open-weight models such as Gemma-2, DeepSeek-R1 distills, and Qwythos. Compare distilled students against their teachers and base models.

References: - Xu et al. (2024): <https://arxiv.org/abs/2402.13116> - Case study: <https://huggingface.co/empero-ai/Qwythos-9B-Claude-Mythos-5-1M>

Topic 13

13. Anatomy of a Harness: How Evaluation Scaffolding Shapes LLM Benchmark Scores

Supervisor: Valerian Foure

An evaluation harness is the fixed code around a model (prompt templates, few-shot formatting, decoding settings, answer parsing, scoring) that turns it into a testable system, so every reported benchmark score is really a model-plus-harness score. Define and review the harness concept, then test harnesses: fix one small open model, vary harness components systematically

References:

- Biderman et al. (2024): <https://arxiv.org/abs/2405.14782>
- <https://github.com/EleutherAI/lm-evaluation-harness>
- Sclar et al. (2024): <https://arxiv.org/abs/2310.11324>

Topic Overview

#	Topic	Supervisor
1	Medical Evidence Agent	JH
2	Gender Bias for Medical Topics	JH
3	Agentic AI for Replicating Empirical Research	MS
4	Jailbreaking Attacks on LLMs	MS
5	SLM-Based Agent Conversion	JR
6	LLMs as Simulated Economic Agents	JR
7	From Paper to Dataset: Agentic AI for Dataset Assembly	EW
8	Benchmarks for Financial LLM Agents	EW
9	Adversarial Attacks and Robustness	JTK
10	Uncertainty in Foundation Models	JTK
11	Do Thumbnails Match Descriptions? Cross-Modal Embedding Alignment on YouTube	VF
12	How Much of a Large Model Can You Distill into a Small One?	VF
13	Anatomy of a Harness: How Evaluation Scaffolding Shapes LLM Benchmark Scores	VF

Next Steps

Next Steps

1. Topic selection and assignment: Google Form
2. Collect and share literature
3. Develop the outline and discuss it with your supervisor
4. Write the seminar paper (LaTeX / Quarto recommended)
5. Prepare the presentation (Quarto / Beamer / PowerPoint)



Note

Questions: Please contact your supervisor or email Prof. Dr. Spindler.

Thank You!



Contact

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Website: <https://www.bwl.uni-hamburg.de/statistik/>

We look forward to an exciting seminar!

Bach, Philipp, Victor Chernozhukov, Malte S. Kurz, and Martin Spindler. 2024. "DoubleML: A Framework for Double Machine Learning in r." *Journal of Statistical Software*.

Bach, Philipp, Malte S. Kurz, Victor Chernozhukov, and Martin Spindler. 2022. "DoubleML - an Object-Oriented Implementation of Double Machine Learning in Python." *Journal of Machine Learning Research*.

Chernozhukov, Victor. 2023. *Causal Inference and Machine Learning*. Draft Manuscript.